LA(1)LR(0)

임기정

잇창명님께 감사드립니다. 다음은 LALR 파서를 생성하는 데에 필요한 정의이다.

(i) $G' \triangleq (N', T', P', S') = (N \uplus \{S'\}, T \uplus \{\$\}, P \uplus \{[S' \to S]\}, S')$.
(ii) $V \triangleq N \uplus T$, $V' \triangleq N' \uplus T'$.
(iii) $L(G) := \{z \in T^* | S \Rightarrow_P^* z\}$, $L(G') := \{z \in T^* | S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^+ z\}$. $L(G) = L(G')$.
(iv) $I \triangleq \{[A \to \alpha \cdot \beta] \mid [A \to \alpha\beta] \in P'\}$.
(v) $\mathrm{Cl}(q) =_\mu q \cup \{[A \to \cdot\omega] | [A \to \omega] \in P', [B \to \alpha_1 \cdot A\alpha_2] \in \mathrm{Cl}(q)\}$ for $q \in \mathcal{P}(I)$.
(vi) $q_0 \triangleq \mathrm{Cl}(\{[S' \to \cdot S]\})$.
(vii) $\mathrm{GOTO}(q, X) \triangleq \mathrm{Cl}(\{[A \to \alpha X \cdot \beta] | [A \to \alpha \cdot X\beta] \in q\})$ for $q \in \mathcal{P}(I)$ and $X \in V$.
(viii) $Q \triangleq \mathrm{PT} \setminus \{\emptyset\}$ where $\mathrm{PT} =_\mu \{q_0\} \cup \{\mathrm{GOTO}(q, X) | q \in \mathrm{PT}, X \in V\}$.
(ix) $\varepsilon : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \;\&\&\; p = q$.
(x) $X\alpha : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \;\&\&\; \alpha : \mathrm{GOTO}(p, X) \to q$.
(xi) $\mathrm{next}(q, X) := \mathrm{GOTO}(q, X)$ for $q \in Q$ and $X \in V$.
(xii) $q_f := \mathrm{GOTO}(q_0, S)$.
(xiii) $\mathrm{reduce}(q, t) := \{[A \to \omega] \mid [A \to \omega\cdot] \in q, t \in \mathrm{LA}(q, [A \to \omega])\}$ for $q \in Q$ and $t \in T'$.
(xiv)

$$\frac{q'' = \mathrm{next}(q', t) \quad q'' \neq \emptyset \quad t \in T}{(\alpha : q \to q', tz) \vdash (\alpha t : q \to q'', z)}\ \text{Shift}$$

(xv)

$$\frac{[A \to \omega] \in \mathrm{reduce}(q', t) \quad q'' = \mathrm{next}(q', A) \quad q'' \neq \emptyset \quad A \in N}{(\alpha\omega : q \to q', tz) \vdash (\alpha A : q \to q'', tz)}\ \mathrm{Reduce}(A \to \omega)$$

(xvi) $L(LRA(G)) := \{z \in T^* | (\varepsilon : q_0 \to q_0, z\$) \vdash^+ (S : q_0 \to q_f, \$)\}$. $L(G) = L(LRA(G))$.
(xvii) For $ts_1 \subseteq \{z \in (T')^* \mid |z| \leq 1\}$ and $ts_2 \subseteq \{z \in (T')^* \mid |z| \leq 1\}$,

$$ts_1 \odot ts_2 := \begin{cases} ts_1, & \text{if } \varepsilon \notin ts_1; \\ (ts_1 \setminus \{\varepsilon\}) \cup ts_2, & \text{otherwise.} \end{cases}$$

(xviii) $\mathrm{First}(A) := \{t | t \in T', \omega \in V^*, A \Rightarrow_{P'}^* t\omega\} \cup \{\varepsilon | A \Rightarrow_{P'}^* \varepsilon\}$ for $A \in N'$.
(xix) $\mathrm{First}(t) := \{t\}$ for $t \in T'$.
(xx) $\mathrm{First}(\varepsilon) := \{\varepsilon\}$, $\mathrm{First}(X\alpha) := \mathrm{First}(X) \odot \mathrm{First}(\alpha)$ for $X \in V'$ and $\alpha \in (V')^*$.

Theorem. (LA) $[A \to \omega\cdot] \in q$인 $q \in Q$, $A \in N'$와 $\omega \in V^*$에 대하여,

$$\mathrm{LA}(q, [A \to \omega]) \triangleq \left\{ t \in T' \;\middle|\; \begin{array}{c} (\alpha\omega : q_0 \to q, tz) \vdash_{\mathrm{Reduce}(A \to \omega)} (\alpha A : q_0 \to q', tz) \\ \&\& \\ (\alpha A : q_0 \to q', tz) \vdash^* (S : q_0 \to q_f, \$) \end{array} \right\}. \tag{1}$$

라 하자. 그러면

$$\mathrm{LA}(q, [A \to \omega]) = \left\{ t \in T' \;\middle|\; S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^* \alpha Az, \alpha\omega : q_0 \to q, \alpha \in V^*, z \in T^*, t \in \mathrm{First}(z\$) \right\} \tag{2}$$

이 성립한다.

이제 $\mathrm{LA}' : \{(q, [A \to \alpha \cdot \beta]) \mid q \in Q, [A \to \alpha \cdot \beta] \in q\} \to \mathcal{P}(\{z \in (T')^* \mid |z| \leq 1\})$를

$$\mathrm{LA}' =_\mu \textbf{propagate}_{\mathrm{LA}'}, \tag{3}$$

로 정의하자. 여기서

$$\frac{}{\$ \in \textbf{propagate}_{\mathrm{LA}'_{\mathrm{previous}}}(q_0, [S' \to \cdot S])}\ \textbf{Base} \tag{4}$$

$$\frac{}{\$ \in \textbf{propagate}_{\mathrm{LA}'_{\mathrm{previous}}}(q_f, [S' \to S\cdot])}\ \textbf{Accept} \tag{5}$$

$$\frac{[A \to \alpha\beta] \in P \quad [A \to \alpha \cdot \beta] \in q \quad \alpha : p \to q \quad [B \to \alpha_1 \cdot A\alpha_2] \in p}{\mathrm{First}(\alpha_2) \odot \mathrm{LA}'_{\mathrm{previous}}(p, [B \to \alpha_1 \cdot A\alpha_2]) \subseteq \textbf{propagate}_{\mathrm{LA}'_{\mathrm{previous}}}(q, [A \to \alpha \cdot \beta])}\ \textbf{Step} \tag{6}$$

이다. 그러면, $q \in Q$이고 $[A \to \omega\cdot] \in q$일 때마다,

$$\mathrm{LA}(q, [A \to \omega]) = \Big\{t \in T' \mid t \in \mathrm{LA}'(q, [A \to \omega\cdot])\Big\} \tag{7}$$

이 성립한다. 이는

$$\mathrm{LA}'(q, [A \to \alpha \cdot \beta]) = \bigcup \left\{\mathrm{First}(z\$) \;\middle|\; S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^* \omega A z \underset{\mathbf{rm}\ P'}{\Longrightarrow} \omega\alpha\beta z, \omega\alpha : q_0 \to q, z \in T^* \right\} \tag{8}$$

이기 때문이다.

Lemma. (lfp) 집합 A가 주어졌다고 하자. 그러면, 함수 $F : \mathcal{P}(A) \to \mathcal{P}(A)$가 단조증가할 때, F의 최소고정점 μ_F가 존재한다.

Proof. A의 부분집합들의 순서열 $\langle X_\alpha \rangle_{\alpha \in \mathbf{Ord}}$을 다음과 같이 초한재귀로 정의하자.

$$X_\alpha \triangleq \bigcup_{\beta < \alpha} F(X_\beta) \quad \text{for}\, \alpha \in \mathbf{Ord}.$$

이제 $\mathcal{P}(A)$의 Hartog 수 $\mathbf{h}(\mathcal{P}(A)) \triangleq \{\zeta \in \mathbf{Ord} \mid \exists i : \zeta \hookrightarrow \mathcal{P}(A)\}$에 대하여

$$\mu_F := X_{\mathbf{h}(\mathcal{P}(A))}$$

로 놓으면, μ_F가 원하던 F의 최소고정점이 된다. □

만약 A가 유한집합이면

$$\mathbf{h}(\mathcal{P}(A)) = 2^{|A|} + 1$$

이다. 우리가 다뤘던 A들은 모두 유한집합이었으므로 구하고자 했던 최소고정점 계산은 항상 수렴하고 F를 최대 $2^{|A|}$번 호출하며 종료한다.